

Cranberries for preventing urinary tract infections.



BACKGROUND: Cranberries have been used widely for several decades for the prevention and treatment of urinary tract infections (UTIs). OBJECTIVES: To assess the effectiveness of cranberry products in preventing UTIs in susceptible populations. SEARCH STRATEGY: We searched MEDLINE, EMBASE, the Cochrane Central Register of Controlled Trials (CENTRAL in The Cochrane Library) and the Internet. We contacted companies involved with the promotion and distribution of cranberry preparations and checked reference lists of review articles and relevant studies. Date of last search: January 2007. SELECTION CRITERIA: All randomised controlled trials (RCTs) or quasi-RCTs of cranberry products for the prevention of UTIs in all populations. DATA COLLECTION AND ANALYSIS: Two authors independently assessed and extracted information. Information was collected on methods, participants, interventions and outcomes (UTIs - symptomatic and asymptomatic, side effects, adherence to therapy). Relative risk (RR) were calculated where appropriate, otherwise a narrative synthesis was undertaken. Quality was assessed using the Cochrane criteria. MAIN RESULTS: Ten studies (n = 1049, five cross-over, five parallel group) were included. Cranberry/cranberry-lingonberry juice versus placebo, juice or water was evaluated in seven studies, and cranberries tablets versus placebo in four studies (one study evaluated both juice and tablets). Cranberry products significantly reduced the incidence of UTIs at 12 months (RR 0.65, 95% CI 0.46 to 0.90) compared with placebo/control. Cranberry products were more effective reducing the incidence of UTIs in women with recurrent UTIs, than elderly men and women or people requiring catheterisation. Six studies were not included in the meta-analyses due to methodological issues or lack of available data. However, only one reported a significant result for the outcome of symptomatic UTIs. Side effects were common in all studies, and dropouts/withdrawals in several of the studies were high. AUTHORS' CONCLUSIONS: There is some evidence that cranberry juice may decrease the number of symptomatic UTIs over a 12 month period, particularly for women with recurrent UTIs. It's effectiveness for other groups is less certain. The large number of dropouts/withdrawals indicates that cranberry juice may not be acceptable over long periods of time. It is not clear what is the optimum dosage or method of administration (e.g. juice, tablets or capsules). Further properly designed studies with relevant outcomes are needed.